

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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CA-1 — ANSWER KEY

Class : V

Max Marks : 70

Subject : Mathematics

Duration : 2 Hr 30 Min

Syllabus: We the Travellers–I • Fractions • Angles as Turns

General Instructions: All questions are compulsory. Read each question carefully before answering.

Section A — 1 Mark Each

1. Which number has the same nearest ten and nearest hundred? (a) 7,126 (b) 7,835 (c) 6,999 (d) 7,030

Ans: (c) 6,999 (nearest ten = 7,000, nearest hundred = 7,000)

2. If $\frac{1}{3} = \frac{2}{x}$, find the value of x.

Ans: $x = 6$

3. Which fraction is the greatest: $\frac{5}{9}$, $\frac{5}{7}$, or $\frac{5}{12}$?

Ans: $\frac{5}{7}$

4. Starting from South, you make a full turn clockwise. Which direction do you end up facing?

Ans: South

5. 3 right angles equal how many quarter turns?

Ans: 3

6. Find the value: $10,000 + 900 + 20 + 4$

Ans: 10,924

7. True or False: An angle can be both acute and a full turn at the same time.

Ans: False

8. How many zeros are there in the number 60,000?

Ans: 4

Section B — 2 Marks Each

9. Choose the digits 1 and 9. Form two 2-digit numbers and find the result of the first subtraction only.

Ans: $91 - 19 = 72$

10. Explain in one line why the repeated-subtraction digit puzzle always ends at 9, no matter which two different digits are chosen.

Ans: Because the difference between any two-digit number and its digit-reversal is always a multiple of 9

11. For the digits 2 and 8, find (difference of digits) $\times$ 9, and check it against $82 - 28$.

Ans: $(8-2) \times 9 = 54$; $82 - 28 = 54$, so they match

12. Compare $\frac{3}{9}$ and $\frac{4}{8}$ using two methods: (i) simplifying each fraction (ii) comparing with $\frac{1}{2}$.

Ans: $\frac{3}{9} = \frac{1}{3} < \frac{1}{2}$; $\frac{4}{8} = \frac{1}{2}$; so $\frac{4}{8} > \frac{3}{9}$ by both methods

13. Give one example of a number whose nearest hundred and nearest thousand are the same, and explain why this happens.

Ans: e.g. 7,030: nearest hundred = 7,000, nearest thousand = 7,000; this happens because the tens/ones part is small enough that both round down to the same value

14. If the minute hand turns $\frac{4}{12}$ of a full turn from 12, how many minutes does this represent, and what is $\frac{4}{12}$ in its simplest form?

Ans: $\frac{4}{12} \times 60 = 20$ minutes; simplest form of $\frac{4}{12}$ is $\frac{1}{3}$

15. Starting from West: turn 3 right angles clockwise, then $\frac{1}{2}$ right angle clockwise, then another $\frac{1}{2}$ right angle clockwise. Find the final direction.

Ans: 3 right angles + $\frac{1}{2}$ + $\frac{1}{2}$ right angle = 4 right angles = 1 full turn; you end up facing West again

16. Find the fraction equivalent to $\frac{5}{9}$ with numerator 25, then compare it with $\frac{25}{54}$.

Ans: $\frac{5}{9} = \frac{25}{45}$; since $\frac{25}{45}$ and $\frac{25}{54}$ have the same numerator, $\frac{25}{45} > \frac{25}{54}$ (smaller denominator), so $\frac{5}{9} > \frac{25}{54}$

17. Ravi's car has been driven 56,987 km and Sheetal's car 67,543 km. Round each distance to the nearest ten-thousand and compare using only the rounded values.

Ans: 56,987 rounds to 60,000; 67,543 rounds to 70,000; by the rounded values, Sheetal's car has been driven more

18. Using the digits 4, 8, 2, 4, 7 (each used exactly once, as given), write the smallest and the largest possible 5-digit numbers.

Ans: Smallest = 24,478; Largest = 87,442

19. The nearest thousand of 45,867 is found, and then 3 is added to it. Find the resulting number and its nearest hundred.

Ans: Nearest thousand of 45,867 = 46,000; +3 = 46,003; nearest hundred of 46,003 = 46,000

20. Find the value of 90 Tens + 83 Ones, and also of 700 Tens + 34 Ones. What is the difference between the two?

Ans: 90 Tens + 83 Ones = 983; 700 Tens + 34 Ones = 7,034; Difference = $7,034 - 983 = 6,051$

21. Four quarter turns are made one after another in the same direction. Name the resulting turn, and explain why it cannot be called an acute, right, obtuse, or straight angle.

Ans: Four quarter turns make a full turn (360 degrees); it returns exactly to the starting position, so it is not classified as acute, right, obtuse, or straight — those describe angles less than or equal to a straight angle

22. Arrange the fractions $\frac{7}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{1}{8}$ from smallest to largest.

Ans: $\frac{1}{8} < \frac{3}{8} < \frac{5}{8} < \frac{7}{8}$

Section C — 3 Marks Each

23. A cyclist travels at 15 km/hr and a motorbike at 50 km/hr. In 3 hours, how much farther does the motorbike travel than the cyclist? Show your working.

Ans: Cyclist: $15 \times 3 = 45$ km; Motorbike: $50 \times 3 = 150$ km; Difference = $150 - 45 = 105$ km

24. Rearrange the digits of 48,247 (using all 5 digits) to form the smallest possible 5-digit number and the largest possible 5-digit number. State both.

Ans: Smallest = 24,478; Largest = 87,442

25. A school has 461 girls and 439 boys. If the school uses exactly 24 Minibuses (25 seats each) first, how many students remain, and how many Boats (20 seats each) are needed to carry them with no seats empty?

Ans: Total = 900; Minibuses carry $24 \times 25 = 600$; Remaining = $900 - 600 = 300$ students; Boats needed = $300 / 20 = 15$ boats exactly

26. Compare $12/9$ and $4/5$ using 1 (a whole) as a reference. If these represent pizzas eaten, which represents 'more than one full pizza'?

Ans: $12/9 > 1$ (more than a whole) and $4/5 < 1$; so $12/9$ represents more than a full pizza

27. A giant wheel passenger completes 5 quarter turns in the same direction from the starting point. How many full turns and extra turns is this? Name the angle formed by the extra turn.

Ans: 5 quarter turns = 1 full turn + 1 extra quarter turn; the extra $1/4$ turn forms a right angle

28. Find the missing values: 37,608 = ____ Ten Thousand + 17 Thousand + ____ Hundreds + 8 Ones. Show your reasoning.

Ans: 17 Thousand = 1 Ten Thousand + 7 Thousand; so total Ten Thousand needed = 2 (since 2 TTh + 17 Th = $20,000 + 17,000 = 37,000$); remaining hundreds = 6; Answer: 2 Ten Thousand, 6 Hundreds

Section D — 4 Marks Each (Word / Application Problems)

29. Town populations: Town 1 = 65,232; Town 3 = 56,380; Town 5 = 45,858; Town 6 = 66,540. (a) Arrange in descending order. (b) Round each to the nearest thousand. (c) State which towns' populations round up and which round down, with reasons.

Ans: (a) $66,540 > 65,232 > 56,380 > 45,858$. (b) 67,000; 65,000; 56,000; 46,000. (c) Town 1 (65,232) rounds down since hundreds digit $2 < 5$; Town 3 (56,380) rounds down since hundreds digit $3 < 5$; Town 5 (45,858) rounds up since hundreds digit $8 \geq 5$; Town 6 (66,540) rounds up since hundreds digit $5 \geq 5$

30. Choose two digits whose difference is 5. Form the two possible 2-digit numbers, subtract, and continue the process (using the digits of each new result) until you reach a 1-digit number. Show at least 2 subtraction steps.

Ans: Example with digits 1 and 6: $61 - 16 = 45$; then $54 - 45 = 9$. Final 1-digit number = 9 (any valid pair with difference 5 works)

Section E — 2 Marks Each (From Your Textbook)

31. Take the digits 3 and 7. Form the two possible 2-digit numbers, subtract the smaller from the bigger, and repeat the process with the digits of each new result until you reach a 1-digit number. Show all your steps.

Ans: $73 - 37 = 36$; $63 - 36 = 27$; $72 - 27 = 45$; $54 - 45 = 9$. Final digit = 9

32. Rearrange the digits of 48,247 (using all 5 digits) to form the largest possible 5-digit number.

Ans: 87,442

33. Give one valid combination of exactly 7 fraction-kit pieces (from halves, thirds, quarters, fifths, sixths, eighths, or twelfths) that add up to a whole, and show your working.

Ans: 1 piece of $\frac{1}{2}$ + 6 pieces of $\frac{1}{12}$ = $\frac{1}{2}$ + $\frac{6}{12}$ = $\frac{1}{2}$ + $\frac{1}{2}$ = 1 (7 pieces total)

34. In the 'pile of pebbles' game, each pile starts with 7 pebbles; players alternate taking any number of pebbles from one pile, and whoever takes the last pebble wins. If it is your turn with piles of (2, 2) remaining, describe and explain a strategy to try to win.

Ans: Accept any well-reasoned strategy describing how to control the pebble counts (e.g. matching/mirroring the opponent's move in the other pile) to try to force the opponent into taking the last pebble or being left without a good move; the textbook treats this as an open exploration.